

One, two, three, we're...

Counting

Basic Counting Principles

Counting problems are of the following kind:

"How many different 8-letter passwords are there?"

"How many possible ways are there to pick 11 soccer players out of a 20-player team?"

Most importantly, counting is the basis for computing probabilities of discrete events.

("What is the probability of winning the lottery?")

Basic Counting Principles

The sum rule:

If a task can be done in n_1 ways and a second task in n_2 ways, and if these two tasks cannot be done at the same time, then there are $n_1 + n_2$ ways to do either task.

Example:

The department will award a free computer to either a CS student or a CS professor. How many different choices are there, if there are 530 students and 15 professors?

There are $530 + 15 = 545$ choices.

Basic Counting Principles

Generalized sum rule:

If we have tasks $T_1, T_2, \dots, T_m$ that can be done in $n_1, n_2, \dots, n_m$ ways, respectively, and no two of these tasks can be done at the same time, then there are $n_1 + n_2 + \dots + n_m$ ways to do one of these tasks.

Basic Counting Principles

The product rule:

Suppose that a procedure can be broken down into two successive tasks. If there are n_1 ways to do the first task and n_2 ways to do the second task after the first task has been done, then there are $n_1 n_2$ ways to do the procedure.

Basic Counting Principles

Example:

How many different license plates are there that containing exactly three English letters ?

Solution:

If repetition of letter is allowed then the answer is right but different answer for non-repetition($26 \cdot 25 \cdot 24$)

There are 26 possibilities to pick the first letter, then 26 possibilities for the second one, and 26 for the last one.

So there are $26 \cdot 26 \cdot 26 = 17576$ different license plates.

Basic Counting Principles

Generalized product rule:

If we have a procedure consisting of sequential tasks $T_1, T_2, \dots, T_m$ that can be done in $n_1, n_2, \dots, n_m$ ways, respectively, then there are $n_1 \cdot n_2 \cdot \dots \cdot n_m$ ways to carry out the procedure.

Basic Counting Principles

The sum and product rules can also be phrased in terms of **set theory**.

Sum rule: Let $A_1, A_2, \dots, A_m$ be disjoint sets. Then the number of ways to choose any element from one of these sets is $|A_1 \cup A_2 \cup \dots \cup A_m| = |A_1| + |A_2| + \dots + |A_m|$.

Product rule: Let $A_1, A_2, \dots, A_m$ be finite sets. Then the number of ways to choose one element from each set in the order $A_1, A_2, \dots, A_m$ is $|A_1 \times A_2 \times \dots \times A_m| = |A_1| \cdot |A_2| \cdot \dots \cdot |A_m|$.

The Pigeonhole Principle

The pigeonhole principle: If $(k + 1)$ or more objects are placed into k boxes, then there is **at least** one box containing two or more of the objects.

Example 1: If there are 11 players in a soccer team that wins 12-0, there must be at least one player in the team who scored at least twice.

Example 2: If you have 6 classes from Monday to Friday, there must be at least one day on which you have at least two classes.

The Pigeonhole Principle

The generalized pigeonhole principle: If N objects are placed into k boxes, then there is **at least** one box containing at least $\lceil N/k \rceil$ of the objects.

Example 1: In our 60-student class, at least 12 students will get the same letter grade (A, B, C, D, or F).

Permutations and Combinations

How many ways are there to pick a set of 3 people from a group of 6?

There are 6 choices for the first person, 5 for the second one, and 4 for the third one, so there are $6 \cdot 5 \cdot 4 = 120$ ways to do this.

This is not the correct result!

For example, picking person C, then person A, and then person E leads to the **same group** as first picking E, then C, and then A.

However, these cases are counted **separately** in the above equation.

Permutations and Combinations

So how can we compute how many different subsets of people can be picked (that is, we want to disregard the order of picking) ?

To find out about this, we need to look at **permutations**.

A **permutation** of a set of distinct objects is an ordered arrangement of these objects.

An ordered arrangement of r elements of a set is called an **r -permutation**.

Permutations and Combinations

Example: Let $S = \{1, 2, 3\}$.

The arrangement 3, 1, 2 is a permutation of S .

The arrangement 3, 2 is a 2-permutation of S .

The number of r -permutations of a set with n distinct elements is denoted by $P(n, r)$.

We can calculate $P(n, r)$ with the product rule:

$$P(n, r) = n \cdot (n - 1) \cdot (n - 2) \cdot \dots \cdot (n - r + 1).$$

(n choices for the first element, $(n - 1)$ for the second one, $(n - 2)$ for the third one...)

Permutations and Combinations

Example:

$$\begin{aligned} P(8, 3) &= 8 \cdot 7 \cdot 6 = 336 \\ &= (8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1) / (5 \cdot 4 \cdot 3 \cdot 2 \cdot 1) \end{aligned}$$

General formula:

$$P(6,3) = 6 \cdot 5 \cdot 4 \cdot 3!$$

$$P(n, r) = n! / (n - r)!$$

Knowing this, we can return to our initial question:
How many ways are there to pick a set of 3 people from a group of 6 (disregarding the order of picking)?

Permutations and Combinations

An **r-combination** of elements of a set is an unordered selection of r elements from the set. Thus, an r -combination is simply a subset of the set with r elements.

Example: Let $S = \{1, 2, 3, 4\}$.

Then $\{1, 3, 4\}$ is a 3-combination from S .

The number of r -combinations of a set with n distinct elements is denoted by $C(n, r)$.

Example: $C(4, 2) = 6$, since, for example, the 2-combinations of a set $\{1, 2, 3, 4\}$ are $\{1, 2\}$, $\{1, 3\}$, $\{1, 4\}$, $\{2, 3\}$, $\{2, 4\}$, $\{3, 4\}$.

Permutations and Combinations

How can we calculate $C(n, r)$?

Consider that we can obtain the r -permutation of a set in the following way:

First, we form all the r -combinations of the set (there are $C(n, r)$ such r -combinations).

Then, we generate all possible orderings in each of these r -combinations (there are $P(r, r)$ such orderings in each case).

Therefore, we have:

$$P(n, r) = C(n, r) \cdot P(r, r)$$

Permutations and Combinations

$$\begin{aligned}C(n, r) &= P(n, r)/P(r, r) \\&= n!/(n-r)!/(r!/(r-r)!) \\&= n!/(r!(n-r)!)\end{aligned}$$

Now we can answer our initial question:

How many ways are there to pick a set of 3 people from a group of 6 (disregarding the order of picking)?

$$C(6, 3) = 6!/(3! \cdot 3!) = 720/(6 \cdot 6) = 720/36 = 20$$

There are 20 different ways, that is, 20 different groups to be picked.

Permutations and Combinations

Example:

A soccer club has 8 female and 7 male members. For today's match, the coach wants to have 6 female and 5 male players on the grass. How many possible configurations are there?

$$\begin{aligned}C(8, 6) \cdot C(7, 5) &= 8!/(6! \cdot 2!) \cdot 7!/(5! \cdot 2!) \\&= 28 \cdot 21 \\&= 588\end{aligned}$$

Combinations

Pascal's Identity:

Let n and k be positive integers with $n \geq k$.
Then $C(n+1, k) = C(n, k-1) + C(n, k)$.

How can this be explained?

What is it good for?

Pascal's Triangle

In Pascal's triangle, each number is the sum of the numbers to its upper left and upper right:



Pascal's Triangle

Since we have $C(n+1, k) = C(n, k-1) + C(n, k)$ and $C(0, 0) = 1$, we can use Pascal's triangle to simplify the computation of $C(n, k)$:

